

Multi-objective Control of a Vehicle with Triple Trailers

Kazuo TANAKA and Shigeki HORI

Department of Mechanical Engineering and Intelligent Systems

The University of Electro-Communications

1-5-1 Chofugaoka, Chofu, Tokyo 182-8585 Japan

Tel. & Fax. +81-424-43-5425

Email ktanaka@mce.uec.ac.jp

Hua O. Wang

Department of Electrical and Computer Engineering

Duke University

Durham, NC 27708-0291 USA

Tel. (919) 660-5273

Fax. (919) 660-5293

Email hua@ee.duke.edu

Corresponding Author

Kazuo Tanaka

Department of Mechanical Engineering and Intelligent Systems

The University of Electro-Communications

1-5-1 Chofugaoka, Chofu, Tokyo 182-8585 Japan

Tel. & Fax. +81-424-43-5425

Email ktanaka@mce.uec.ac.jp

Abstract

We consider the backing up control of a vehicle with triple trailers using a model-based fuzzy control methodology. To begin with, the vehicle model is represented by a Takagi-Sugeno fuzzy model. Then we employ the so-called "parallel distributed compensation" design to arrive at a controller that guarantees the stability of the closed-loop system consisted of the fuzzy model and controller. The control design problem is cast in terms of linear matrix inequalities (LMIs). In addition to stability, control performance considerations such as decay rate, constraints on input and output and disturbance rejection are incorporated in the LMI conditions. In application to the vehicle with triple trailers setup, we utilize these LMI conditions to explicitly avoid the saturation of the steering angle and the jack-knife phenomenon in the control design. Both simulation and experimental results are presented. Our results demonstrate that the fuzzy controller effectively achieves the backing up control of the vehicle with triple trailers while avoiding the saturation of the actuator and "jack-knife" phenomenon.

Keywords : fuzzy control, linear matrix inequalities, fuzzy modeling, the relaxed stability condition, decay rate, constraints on control input and output, disturbance rejection, nonlinear systems

1 Introduction

The backing up control of "trailer-truck", i.e., a vehicle with a trailer, has been used as a testbed for a variety of control design methods, e.g., [1, 2, 3]. In order to successfully back

up the trailer-truck, the so-called "jack-knife" phenomenon should be avoided. Intelligent control techniques [4]-[8] have been applied to this control problem. Since the papers [4]-[7] dealt with empirical and non-systematic designs, even stability of the closed-loop system was not analyzed. The paper [8] theoretically discussed an excellent design method for this problem. However, they reported only simulation results of a vehicle with a trailer. On the other hand, Sampei reported experimental results of a vehicle with a trailer [9] and double trailer [10] via a systematic nonlinear control technique based on exact linearization.

The first purpose of this paper is to achieve backing control of a vehicle with triple trailers in experiments. The triple trailers case is much more difficult than the single or double trailers case. The single trailer case exists only two jack-knife positions. The triple trailers case exists eight jack-knife positions. As far as we know, experimental results for the triple trailers case have not been reported in the literature. The second purpose is to realize a multi-objective control design from the practical points of view. The backing control experiments require many practical constraints such as avoidance of steering angle saturation, jack-knife phenomenon, measurement noise rejection, and so on. To succeed in the triple trailers backing control experiments, we need a multi-objective control design satisfying these practical constraints. The LMI-based fuzzy control presented here achieves a multi-objective control design satisfying these practical constraints. One of the authors designed a stable fuzzy controller for the single trailer case [11]. However, the stable feedback gains were found by a trial and error type of procedure. The controller design presented in this paper is more effective.

Recently, fuzzy control has been widely and successfully applied to nonlinear processes,

e.g., [17]-[28]. Our approach is completely different from the typical fuzzy control approach. First a Takagi-Sugeno fuzzy model [29] is constructed to describe the nonlinear model of the vehicle with triple trailers. Once the fuzzy model is obtained, the control design is carried out based on the so-called parallel distributed compensation (PDC) scheme [13]-[16]. The design procedure aims at rendering stabilizing controller satisfying performance specifications. The design procedure is systematic and conceptually simple. A key feature of the design framework is that the stability analysis and design problems are cast in terms of linear matrix inequalities (LMIs). In addition to stability, control performance considerations such as decay rate and constraints on input and output are also incorporated in the LMI conditions. The LMI constraints on control input and output depend on initial states. We derive LMI constraints on control input and output that are independent of initial states. Moreover, we derive an LMI condition for disturbance rejection that is related to L_2 gain of the system. In application to the vehicle with triple trailers setup, we utilize these LMI conditions to explicitly avoid the saturation of the steering angle and the jack-knife phenomenon in the control design. Both simulation and experimental results are presented. Our results demonstrate that the fuzzy controller effectively achieves the backing up control of the vehicle with triple trailers while avoiding the saturation of the actuator and "jack-knife" phenomenon. Moreover, the feedback controller guarantees the stability and performance even for disturbance.

Section 2 presents the fuzzy modeling of the vehicle with triple trailers. Section 3 recalls LMI based fuzzy control design conditions. Control design for the vehicle and associated simulation results are presented in Section 4. Experimental results are presented in Section

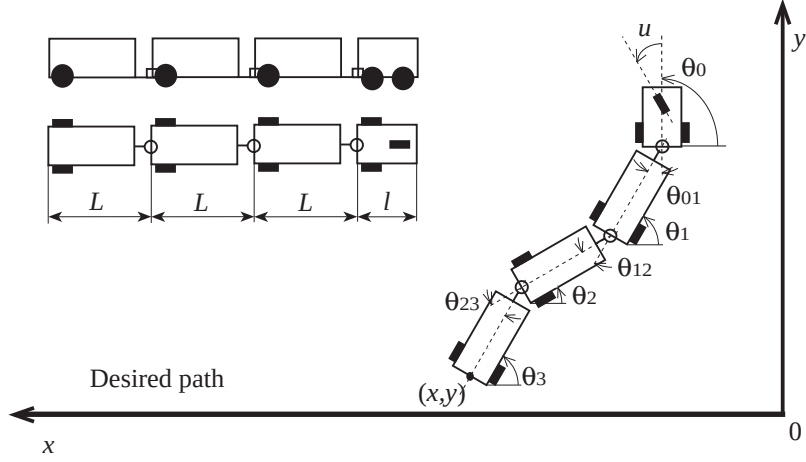


Figure 1: Vehicle model with triple trailers.

5. Concluding remarks are collected in Section 6.

2 Fuzzy Model of a Vehicle with Triple Trailers

Figure 1 shows the coordinate system of a vehicle model with triple trailers. We use the following model in this paper:

$$\theta_0(t+1) = \theta_0(t) + \frac{\nu \cdot \Delta t}{\ell} \tan(u(t)) \quad (1)$$

$$\theta_{01}(t) = \theta_0(t) - \theta_1(t) \quad (2)$$

$$\theta_1(t+1) = \theta_1(t) + \frac{\nu \cdot \Delta t}{L} \sin(\theta_{01}(t)) \quad (3)$$

$$\theta_{12}(t) = \theta_1(t) - \theta_2(t) \quad (4)$$

$$\theta_2(t+1) = \theta_2(t) + \frac{\nu \cdot \Delta t}{L} \sin(\theta_{12}(t)) \quad (5)$$

$$\theta_{23}(t) = \theta_2(t) - \theta_3(t) \quad (6)$$

$$\theta_3(t+1) = \theta_3(t) + \frac{\nu \cdot \Delta t}{L} \sin(\theta_{23}(t)) \quad (7)$$

$$x(t+1) = x(t) + \nu \cdot \Delta t \cos(\theta_{23}(t)) \cos \frac{\theta_3(t+1) + \theta_3(t)}{2} \quad (8)$$

$$y(t+1) = y(t) + \nu \cdot \Delta t \cos(\theta_{23}(t)) \sin \frac{\theta_3(t+1) + \theta_3(t)}{2} \quad (9)$$

where

$\theta_0(t)$: angle of vehicle,

$\theta_i(t)$: angle of i th trailer,

$\theta_{01}(t)$: relative angle between vehicle and first trailer,

$\theta_{ij}(t)$: relative angle between i th trailer and j th trailer,

$x(t)$: horizontal position of rear end of third trailer,

$y(t)$: vertical position of rear end of third trailer,

$u(t)$: steering angle.

In the simulation and experiment studies of this paper we have

$$\ell = 0.087[m], \quad L = 0.130[m], \quad \nu = -0.10[m/sec.], \quad \Delta t = 0.5[sec.],$$

where ℓ is the length of the vehicle, L is the length of the trailer, Δt is the sampling time, and ν is the constant speed of the backward movement. For $\theta_{01}(t)$, $\theta_{12}(t)$ and $\theta_{23}(t)$, 90 [deg.] and -90 [deg.] correspond to eight "jack-knife" positions. Figure 2 shows the typical process of the jack-knife phenomenon for the single trailer case. The jack-knife phenomenon can not be avoided if the steering is not controlled during the backward movement. To succeed in the backing control, we need to avoid the jack-knife phenomenon. The simulation results will show that an LMI constraint on the control output is useful for avoiding the jack-knife phenomenon.

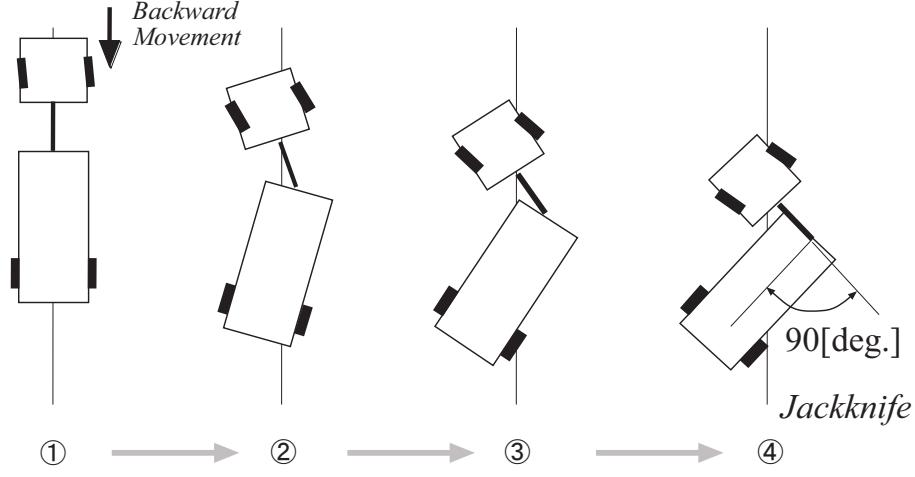


Figure 2: Jack-knife phenomenon.

The control purpose is to back the vehicle into the straight line ($y = 0$) without any forward movement, that is,

$$\theta_{01}(t) \rightarrow 0, \theta_{12}(t) \rightarrow 0, \theta_{23}(t) \rightarrow 0, \theta_3(t) \rightarrow 0, y(t) \rightarrow 0.$$

We construct the following fuzzy model from the model. The details of fuzzy modeling will be described in Appendix.

Rule 1 : If $p(t)$ is "about 0 [rad.]",

$$\text{then } \mathbf{x}(t+1) = \mathbf{A}_1 \mathbf{x}(t) + \mathbf{B}_1 u(t), \quad (10)$$

Rule 2 : If $p(t)$ is "about π [rad.]" or " $-\pi$ [rad.]",

$$\text{then } \mathbf{x}(t+1) = \mathbf{A}_2 \mathbf{x}(t) + \mathbf{B}_2 u(t),$$

where

$$p(t) = \theta_3(t) + \frac{\nu \cdot \Delta t}{2L} \theta_{23}(t),$$

$$\begin{aligned}
\mathbf{x}(t) &= \begin{bmatrix} \theta_{01}(t) & \theta_{12}(t) & \theta_{23}(t) & \theta_3(t) & y(t) \end{bmatrix}^T, \\
\mathbf{A}_1 &= \begin{bmatrix} 1 - \frac{\nu \cdot \Delta t}{L} & 0 & 0 & 0 & 0 \\ \frac{\nu \cdot \Delta t}{L} & 1 - \frac{\nu \cdot \Delta t}{L} & 0 & 0 & 0 \\ 0 & \frac{\nu \cdot \Delta t}{L} & 1 - \frac{\nu \cdot \Delta t}{L} & 0 & 0 \\ 0 & 0 & \frac{\nu \cdot \Delta t}{L} & 1 & 0 \\ 0 & 0 & \frac{(\nu \cdot \Delta t)^2}{2L} & \nu \cdot \Delta t & 1 \end{bmatrix}, \\
\mathbf{B}_1 &= \begin{bmatrix} \frac{\nu \cdot \Delta t}{\ell} \\ 0 \\ 0 \\ 0 \end{bmatrix}, \\
\mathbf{A}_2 &= \begin{bmatrix} 1 - \frac{\nu \cdot \Delta t}{L} & 0 & 0 & 0 & 0 \\ \frac{\nu \cdot \Delta t}{L} & 1 - \frac{\nu \cdot \Delta t}{L} & 0 & 0 & 0 \\ 0 & \frac{\nu \cdot \Delta t}{L} & 1 - \frac{\nu \cdot \Delta t}{L} & 0 & 0 \\ 0 & 0 & \frac{\nu \cdot \Delta t}{L} & 1 & 0 \\ 0 & 0 & \frac{g \cdot (\nu \cdot \Delta t)^2}{2L} & g \cdot \nu \cdot \Delta t & 1 \end{bmatrix}, \\
\mathbf{B}_2 &= \begin{bmatrix} \frac{\nu \cdot \Delta t}{\ell} \\ 0 \\ 0 \\ 0 \end{bmatrix}.
\end{aligned}$$

The overall fuzzy model is inferred as

$$\begin{aligned}
\mathbf{x}(t+1) &= \frac{\sum_{i=1}^r w_i(p(t)) \{ \mathbf{A}_i \mathbf{x}(t) + \mathbf{B}_i u(t) \}}{\sum_{i=1}^r w_i(p(t))} \\
&= \sum_{i=1}^r h_i(p(t)) \{ \mathbf{A}_i \mathbf{x}(t) + \mathbf{B}_i u(t) \}, \tag{11}
\end{aligned}$$

where

$$h_i(p(t)) = \frac{w_i(p(t))}{\sum_{i=1}^r w_i(p(t))}.$$

r denotes the number of rules. In this case, $r = 2$. Figure 3 shows the membership functions "about 0 [rad.]" and "about π [rad.] or $-\pi$ [rad.]".

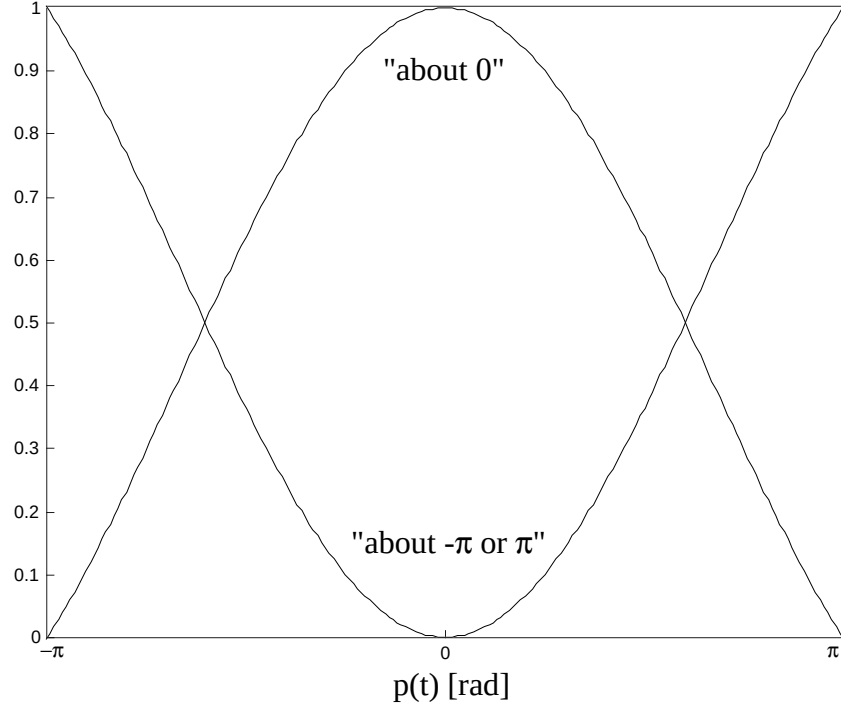


Figure 3: Membership functions.

Next, we design a fuzzy controller for the fuzzy model (10) of the vehicle. The parallel distributed compensation (PDC) introduced in [13]-[16] is employed to arrive at a fuzzy controller. The main idea of the PDC design is to associate a compensatory for each rule of the fuzzy model. The fuzzy controller shares the same fuzzy rules with the fuzzy model, i.e., for Rule 1 and Rule 2 in the fuzzy model (10), we have Control rule 1 and Control rule 2 in the control design.

Control rule 1 : If $p(t)$ is "about 0 [rad.]",

$$\text{then } u(t) = -\mathbf{F}_1 \mathbf{x}(t), \quad (12)$$

Control rule 2 : If $p(t)$ is "about π [rad.]" or $-\pi$ [rad.]",

$$\text{then } u(t) = -\mathbf{F}_2 \mathbf{x}(t).$$

The fuzzy control rules have linear state feedback laws in the consequent parts. The overall fuzzy controller is given by

$$u(t) = -\frac{\sum_{i=1}^r w_i(p(t)) \mathbf{F}_i \mathbf{x}(t)}{\sum_{i=1}^r w_i(p(t))} = -\sum_{i=1}^r h_i(p(t)) \mathbf{F}_i \mathbf{x}(t). \quad (13)$$

r denotes the number of rules. In this case, $r = 2$. The fuzzy controller design problem is to determine the feedback gains $\mathbf{F}_i$ in the consequent parts. Thus, the PDC concept is simple and natural. Typical nonlocal nonlinear control designs would require rather involved procedures.

Substituting (13) into (11), we obtain

$$\mathbf{x}(t+1) = \sum_{i=1}^r \sum_{j=1}^r h_i(p(t)) h_j(p(t)) \{\mathbf{A}_i - \mathbf{B}_i \mathbf{F}_j\} \mathbf{x}(t). \quad (14)$$

Equations (14) can be rewritten as

$$\begin{aligned} \mathbf{x}(t+1) &= \sum_{i=1}^r h_i^2(p(t)) \mathbf{G}_{ii} \mathbf{x}(t) \\ &+ 2 \sum_{i=1}^r \sum_{i < j} h_i(p(t)) h_j(p(t)) \left(\frac{\mathbf{G}_{ij} + \mathbf{G}_{ji}}{2} \right) \mathbf{x}(t), \end{aligned} \quad (15)$$

where $\mathbf{G}_{ij} = \mathbf{A}_i - \mathbf{B}_i \mathbf{F}_j$.

3 Multi-objective Control Design

This section presents a multi-objective control design based on control performance represented in terms of LMIs. We first stated in [14]-[16] that the common $\mathbf{P}$ problem in stable fuzzy controller design can be numerically solved using LMI based technique. The LMI based designs allow us to systematically design PDC fuzzy controllers which satisfy not only

quadratic stability [12]-[16] but also decay rate, constraints on control input and output [30], etc. Section 3.1 recalls the LMI conditions for quadratic stability, decay rate and constraints on control input and output. The simulation results will show that the LMI constraints on the control input and output are useful to explicitly avoid the saturation of the steering angle and the jack-knife phenomenon. To utilize the LMI constraints, the initial states should be known in advance. We derive an LMI condition that is independent of the initial states in Section 3.2. Measurement noise can not be avoided in experiments. In Section 3.3, we derive a disturbance rejection condition that is also represented in terms of LMIs.

3.1 Stability, Decay rate, constraints on control input and output

Theorem 1 [30] *Assume that the number of rules that fire for all t is less than or equal to s , where $1 < s \leq r$. The closed-loop system (15) is stable if there exist a common positive definite matrix $\mathbf{X}$, a common positive semidefinite matrix $\mathbf{Y}$ and $\mathbf{M}_i$ ($i = 1, 2$) satisfying*

$$\begin{bmatrix} \mathbf{X} - (s-1)\mathbf{Y} & \mathbf{X}\mathbf{A}_i^T - \mathbf{M}_i^T\mathbf{B}_i^T \\ \mathbf{A}_i\mathbf{X} - \mathbf{B}_i\mathbf{M}_i & \mathbf{X} \end{bmatrix} > \mathbf{0} \quad (16)$$

for all i and

$$\begin{bmatrix} \mathbf{X} + \mathbf{Y} \\ 1/2(\mathbf{A}_i\mathbf{X} + \mathbf{A}_j\mathbf{X} - \mathbf{B}_i\mathbf{M}_j - \mathbf{B}_j\mathbf{M}_i) \\ 1/2(\mathbf{A}_i\mathbf{X} + \mathbf{A}_j\mathbf{X} - \mathbf{B}_i\mathbf{M}_j - \mathbf{B}_j\mathbf{M}_i)^T \\ \mathbf{X} \end{bmatrix} \geq \mathbf{0} \quad (17)$$

for $i < j$ excepting all the pairs (i, j) such that $h_i(p(t))h_j(p(t)) = 0, \forall t$, where

$$\mathbf{M}_i = \mathbf{F}_i\mathbf{X}.$$

[Remark 2.1]

It is well known that the stability conditions of Theorem 1 can be simplified in the case of common $\mathbf{B}$ ($\mathbf{B}_1 = \dots = \mathbf{B}_r$) [15], [16]. In this paper we consider general design conditions, i.e., not the common $\mathbf{B}$ case, although the fuzzy model of the vehicle shares common $\mathbf{B}$ among the rules.

The above conditions are LMIs in variables $\mathbf{X}$, $\mathbf{Y}$ and $\mathbf{M}_i$. By using convex optimization techniques, we can find a positive definite matrix $\mathbf{X}$, a positive semidefinite matrix $\mathbf{Y}$ and matrices $\mathbf{M}_i$'s satisfying the LMI conditions or determine that no such $\mathbf{X}$, $\mathbf{Y}$ and $\mathbf{M}_i$'s exist. The feedback gains $\mathbf{F}_i$ can be then obtained as

$$\mathbf{F}_i = \mathbf{M}_i \mathbf{X}^{-1} \quad (18)$$

from the solutions, $\mathbf{X}$ and $\mathbf{M}_i$.

Theorem 2 [30] *The largest lower bounded on the decay rate that we can find using a quadratic Lyapunov function $V(\mathbf{x}(t)) = \mathbf{x}(t)^T \mathbf{X}^{-1} \mathbf{x}(t)$ can be found by solving the following GEVP (generalized eigenvalue minimization problem) in $\mathbf{X}$, $\mathbf{Y}$ and β , where $\mathbf{Y}$ is a positive semidefinite matrix and $0 \leq \beta < 1$:*

$$\underset{\mathbf{X}, \mathbf{Y}, \mathbf{M}_1, \dots, \mathbf{M}_r}{\text{minimize}} \quad \beta$$

subject to

$$\mathbf{X} > \mathbf{0}, \quad \mathbf{Y} \geq \mathbf{0},$$

$$\begin{bmatrix} \beta \mathbf{X} - (s-1)\mathbf{Y} & \mathbf{X}\mathbf{A}_i^T - \mathbf{M}_i^T \mathbf{B}_i^T \\ \mathbf{A}_i \mathbf{X} - \mathbf{B}_i \mathbf{M}_i & \beta \mathbf{X} \end{bmatrix} > \mathbf{0}, \quad (19)$$

for all i and

$$\begin{bmatrix} \beta \mathbf{X} + \mathbf{Y} \\ \frac{1}{2}(\mathbf{A}_i \mathbf{X} + \mathbf{A}_j \mathbf{X} - \mathbf{B}_i \mathbf{M}_j - \mathbf{B}_j \mathbf{M}_i) \\ \frac{1}{2}(\mathbf{A}_i \mathbf{X} + \mathbf{A}_j \mathbf{X} - \mathbf{B}_i \mathbf{M}_j - \mathbf{B}_j \mathbf{M}_i)^T \\ \mathbf{X} \end{bmatrix} \geq \mathbf{0}, \quad (20)$$

for $i < j$ excepting all the pairs (i, j) such that $h_i(p(t))h_j(p(t)) = 0, \forall t$, where

$$\mathbf{M}_i = \mathbf{F}_i \mathbf{X}.$$

Note that inequalities (19) and (20) are reduced to the conditions of Theorem 1 when $\beta = 1$.

Constraints on the control input and output can be also represented in terms of LMIs.

Theorem 3 [30] Assume that the initial state $\mathbf{x}(0)$ is known. The constraint $\|u(t)\|_2 \leq \mu$ is enforced at all times $t \geq 0$ if the LMIs

$$\begin{bmatrix} 1 & \mathbf{x}^T(0) \\ \mathbf{x}(0) & \mathbf{X} \end{bmatrix} \geq \mathbf{0}, \quad (21)$$

$$\begin{bmatrix} \mathbf{X} & \mathbf{M}_i^T \\ \mathbf{M}_i & \mu^2 \mathbf{I} \end{bmatrix} \geq \mathbf{0} \quad (22)$$

hold, where $\mathbf{X} > \mathbf{0}$ and $\mathbf{M}_i = \mathbf{F}_i \mathbf{X}$.

Theorem 4 Assume that the initial state $\mathbf{x}(0)$ is known. The constraint $\|\theta_{01}(t)\| \leq \lambda_1$, $\|\theta_{12}(t)\| \leq \lambda_2$ and $\|\theta_{23}(t)\| \leq \lambda_3$ are enforced at all times $t \geq 0$ if the LMIs

$$\begin{bmatrix} 1 & \mathbf{x}^T(0) \\ \mathbf{x}(0) & \mathbf{X} \end{bmatrix} \geq \mathbf{0}, \quad (23)$$

$$\begin{bmatrix} \mathbf{X} & \mathbf{X}\mathbf{d}_1^T \\ \mathbf{d}_1\mathbf{X} & \lambda_1^2\mathbf{I} \end{bmatrix} \geq \mathbf{0}, \quad (24)$$

$$\begin{bmatrix} \mathbf{X} & \mathbf{X}\mathbf{d}_2^T \\ \mathbf{d}_2\mathbf{X} & \lambda_2^2\mathbf{I} \end{bmatrix} \geq \mathbf{0}, \quad (25)$$

$$\begin{bmatrix} \mathbf{X} & \mathbf{X}\mathbf{d}_3^T \\ \mathbf{d}_3\mathbf{X} & \lambda_3^2\mathbf{I} \end{bmatrix} \geq \mathbf{0} \quad (26)$$

hold, where $\mathbf{X} > \mathbf{0}$. In the triple trailer case, we can select $\theta_{01}(t)$, $\theta_{12}(t)$ and $\theta_{23}(t)$ as outputs, i.e.,

$$\begin{aligned} \theta_{01}(t) &= \mathbf{d}_1\mathbf{x}(t) = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \theta_{01}(t) \\ \theta_{12}(t) \\ \theta_{23}(t) \\ \theta_3(t) \\ y(t) \end{bmatrix}, \\ \theta_{12}(t) &= \mathbf{d}_2\mathbf{x}(t) = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \theta_{01}(t) \\ \theta_{12}(t) \\ \theta_{23}(t) \\ \theta_3(t) \\ y(t) \end{bmatrix}, \\ \theta_{23}(t) &= \mathbf{d}_3\mathbf{x}(t) = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} \theta_{01}(t) \\ \theta_{12}(t) \\ \theta_{23}(t) \\ \theta_3(t) \\ y(t) \end{bmatrix}. \end{aligned}$$

(proof)

We derive the condition (24).

From $\|\theta_{01}(t)\| \leq \lambda_1$,

$$\theta_{01}^T\theta_{01}(t) = \mathbf{x}^T(t)\mathbf{d}_1^T\mathbf{d}_1\mathbf{x}(t) \leq \lambda_1^2.$$

Therefore,

$$\frac{1}{\lambda_1^2}\mathbf{x}^T(t)\mathbf{d}_1^T\mathbf{d}_1\mathbf{x}(t) \leq 1.$$

In the same way as in the proof of Theorem 3 , we have

$$\frac{1}{\lambda_1} \mathbf{x}^T(t) \mathbf{d}_1^T \mathbf{d}_1 \mathbf{x}(t) \leq \mathbf{x}^T(t) \mathbf{X}^{-1} \mathbf{x}(t).$$

The above inequality is

$$\mathbf{x}^T(t) \left(\frac{1}{\lambda_1^2} \mathbf{d}_1^T \mathbf{d}_1 - \mathbf{X}^{-1} \right) \mathbf{x}(t) \leq 0.$$

Therefore, we have

$$\mathbf{X} - \frac{1}{\lambda_1^2} \mathbf{X} \mathbf{d}_1^T \mathbf{d}_1 \mathbf{X} \geq 0.$$

Inequality (24) can be then obtained from the above inequality. We obtain the LMI conditions (25) and (26) in the same fashion. (Q.E.D.)

3.2 Independence condition for initial states

The above LMI design conditions for input and output constraints depend on the initial states of the system. This means that the feedback gains $\mathbf{F}_i$ must be again determined using the above LMIs if the initial states $\mathbf{x}(0)$ changes. This is a disadvantage point of using the LMI on the control input and output. We modify the LMI constraints on the control input and output, where $\mathbf{x}(0)$ is unknown, but the upper bound ϕ of $\|\mathbf{x}(0)\|$ is known, i.e., $\|\mathbf{x}(0)\| \leq \phi$. We can select a large number as ϕ even if $\mathbf{x}(0)$ is unknown. Of course, a very large number leads to conservative results.

The modified LMI is accomplished by the following result.

Theorem 5 *Assume that $\|\mathbf{x}(0)\| \leq \phi$, where $\mathbf{x}(0)$ is unknown but the upper bound ϕ is known. Then,*

$$\mathbf{x}^T(0) \mathbf{X}^{-1} \mathbf{x}(0) \leq 1 \tag{27}$$

if

$$\phi^2 \mathbf{I} \leq \mathbf{X}, \quad (28)$$

where $\mathbf{X} = \mathbf{P}^{-1}$.

(proof)

From (28),

$$\mathbf{X}^{-1} \leq \frac{1}{\phi^2} \mathbf{I}.$$

Therefore,

$$\mathbf{x}^T(0) \mathbf{X}^{-1} \mathbf{x}(0) \leq \frac{1}{\phi^2} \mathbf{x}^T(0) \mathbf{x}(0) \leq 1.$$

Now, we arrive at (27). Note that (27) is equivalent to (21). (Q.E.D.)

3.3 Disturbance rejection

This section presents a fuzzy controller design to reject the disturbance for the following

Takagi-Sugeno fuzzy models:

$$\mathbf{x}(t+1) = \sum_{i=1}^r h_i(p(t)) \{ \mathbf{A}_i \mathbf{x}(t) + \mathbf{B}_i \mathbf{u}(t) + \mathbf{E}_i \mathbf{v}(t) \}, \quad (29)$$

$$\mathbf{y}(t) = \sum_{i=1}^r h_i(p(t)) \mathbf{C}_i \mathbf{x}(t), \quad (30)$$

where $\mathbf{v}(t)$ is the disturbance. The fuzzy model (10) has the common $\mathbf{B}$ matrix ($\mathbf{B}_1 = \mathbf{B}_2$)

and the single input $u(t)$. However, we discuss a more general case described in (29) and

(30). The disturbance rejection can be realized by minimizing γ subject to

$$\sup_{\|\mathbf{v}(t)\|_2 \neq 0} \frac{\|\mathbf{y}(t)\|_2}{\|\mathbf{v}(t)\|_2} \leq \gamma. \quad (31)$$

Theorem 6 *The feedback gains $\mathbf{F}_i$ that stabilize the fuzzy model and minimize γ in (31) can be obtained by solving the following LMIs.*

$$\begin{aligned}
& \underset{\mathbf{X}, \mathbf{M}_1, \dots, \mathbf{M}_r}{\text{minimize}} \quad \gamma^2 \\
& \text{subject to} \\
& \mathbf{X} > \mathbf{0} \\
& \left[\begin{array}{ccc} \mathbf{X} & * & * \\ \mathbf{0} & \gamma^2 \mathbf{I} & * \\ \frac{1}{2}(\mathbf{A}_i \mathbf{X} - \mathbf{B}_i \mathbf{M}_j + \mathbf{A}_j \mathbf{X} - \mathbf{B}_j \mathbf{M}_i) & \frac{1}{2}(\mathbf{E}_i + \mathbf{E}_j) & \mathbf{X} \\ \frac{1}{2}(\mathbf{C}_i + \mathbf{C}_j) \mathbf{X} & \mathbf{0} & \mathbf{0} \end{array} \right] \geq \mathbf{0} \quad i \leq j \quad (32)
\end{aligned}$$

for $i \leq j$ excepting the pairs (i, j) such that $h_i(p(t))h_j(p(t)) = 0, \forall t$, where $\mathbf{X} > \mathbf{0}$, $\mathbf{M}_i = \mathbf{F}_i \mathbf{X}$ and $*$ denotes the transposed elements (matrices) for symmetric positions.

(proof)

Suppose there exists a quadratic function $V(\mathbf{x}(t)) = \mathbf{x}^T(t) \mathbf{P} \mathbf{x}(t)$, $\mathbf{P} > \mathbf{0}$ and $\gamma \geq 0$ such that for all t ,

$$\Delta V(\mathbf{x}(t)) + \mathbf{y}^T(t) \mathbf{y}(t) - \gamma^2 \mathbf{v}^T(t) \mathbf{v}(t) \leq 0, \quad (33)$$

for (29) and (30). From (33), we obtain

$$\sum_{t=0}^T \left\{ \Delta V(\mathbf{x}(t)) + \mathbf{y}^T(t) \mathbf{y}(t) - \gamma^2 \mathbf{v}^T(t) \mathbf{v}(t) \right\} \leq 0.$$

By assuming that initial condition $\mathbf{x}(0) = \mathbf{0}$, we obtain

$$V(\mathbf{x}(T)) + \sum_{t=0}^T \left(\mathbf{y}^T(t) \mathbf{y}(t) - \gamma^2 \mathbf{v}^T(t) \mathbf{v}(t) \right) \leq 0. \quad (34)$$

Since $V(\mathbf{x}(T)) \geq 0$, this implies

$$\frac{\|\mathbf{y}(t)\|_2}{\|\mathbf{v}(t)\|_2} \leq \gamma.$$

Therefore the L_2 gain of the fuzzy model is less than γ if (33) holds. We derive the LMI condition (32) from (33).

$$\begin{aligned}
& \gamma^2 \mathbf{v}^T(t) \mathbf{v}(t) - \mathbf{y}^T(t) \mathbf{y}(t) - \Delta V(\mathbf{x}(t)) \\
= & \gamma^2 \mathbf{v}^T(t) \mathbf{v}(t) - \mathbf{x}^T(t) \sum_{i=1}^r h_i(p(t)) \mathbf{C}_i^T \sum_{i=1}^r h_i(p(t)) \mathbf{C}_i \mathbf{x}(t) \\
& - \left\{ \sum_{i=1}^r \sum_{j=1}^r h_i(p(t)) h_j(p(t)) (\mathbf{A}_i - \mathbf{B}_i \mathbf{F}_j) \mathbf{x}(t) + \sum_{i=1}^r h_i(p(t)) \mathbf{E}_i \mathbf{v}(t) \right\}^T \\
& \times \mathbf{P} \left\{ \sum_{i=1}^r \sum_{j=1}^r h_i(p(t)) h_j(p(t)) (\mathbf{A}_i - \mathbf{B}_i \mathbf{F}_j) \mathbf{x}(t) \right. \\
& \quad \left. + \sum_{i=1}^r h_i(p(t)) \mathbf{E}_i \mathbf{v}(t) \right\} + \mathbf{x}^T(t) \mathbf{P} \mathbf{x}(t) \\
= & \begin{bmatrix} \mathbf{x}^T(t) & \mathbf{v}^T(t) \end{bmatrix} \begin{bmatrix} \mathbf{P} & \mathbf{0} \\ \mathbf{0} & \gamma^2 \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ \mathbf{v}(t) \end{bmatrix} \\
& - \begin{bmatrix} \mathbf{x}^T(t) & \mathbf{v}^T(t) \end{bmatrix} \sum_{i=1}^r \sum_{j=1}^r h_i(p(t)) h_j(p(t)) [\mathbf{A}_i - \mathbf{B}_i \mathbf{F}_j \quad \mathbf{E}_i]^T \\
& \quad \times \mathbf{P} \sum_{i=1}^r \sum_{j=1}^r h_i(p(t)) h_j(p(t)) [\mathbf{A}_i - \mathbf{B}_i \mathbf{F}_j \quad \mathbf{E}_i] \begin{bmatrix} \mathbf{x}(t) \\ \mathbf{v}(t) \end{bmatrix} \\
& \quad - \mathbf{x}^T(t) \sum_{i=1}^r h_i(p(t)) \mathbf{C}_i^T \sum_{i=1}^r h_i(p(t)) \mathbf{C}_i \mathbf{x}(t) \\
= & \begin{bmatrix} \mathbf{x}^T(t) & \mathbf{v}^T(t) \end{bmatrix} \begin{bmatrix} \mathbf{P} - \sum_{i=1}^r h_i(p(t)) \mathbf{C}_i^T \sum_{i=1}^r h_i(p(t)) \mathbf{C}_i & \mathbf{0} \\ \mathbf{0} & \gamma^2 \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ \mathbf{v}(t) \end{bmatrix} \\
& - \begin{bmatrix} \mathbf{x}^T(t) & \mathbf{v}^T(t) \end{bmatrix} \times \sum_{i=1}^r \sum_{j=1}^r h_i(p(t)) h_j(p(t)) [\mathbf{A}_i - \mathbf{B}_i \mathbf{F}_j \quad \mathbf{E}_i]^T \\
& \quad \times \mathbf{P} \sum_{i=1}^r \sum_{j=1}^r h_i(p(t)) h_j(p(t)) [\mathbf{A}_i - \mathbf{B}_i \mathbf{F}_j \quad \mathbf{E}_i] \begin{bmatrix} \mathbf{x}(t) \\ \mathbf{v}(t) \end{bmatrix} \geq \mathbf{0} \tag{35}
\end{aligned}$$

From Schur complement, (35) is equivalent to

$$\begin{bmatrix} \mathbf{P} - \sum_{i=1}^r h_i(p(t)) \mathbf{C}_i^T \sum_{i=1}^r h_i(p(t)) \mathbf{C}_i & \mathbf{0} \\ \mathbf{0} & \gamma^2 \mathbf{I} \\ \sum_{i=1}^r \sum_{j=1}^r h_i(p(t)) h_j(p(t)) (\mathbf{A}_i - \mathbf{B}_i \mathbf{F}_j) & \sum_{i=1}^r h_i(p(t)) \mathbf{E}_i \end{bmatrix}$$

$$\left[\begin{array}{c} \sum_{i=1}^r \sum_{j=1}^r h_i(p(t)) h_j(p(t)) \{ \mathbf{A}_i - \mathbf{B}_i \mathbf{F}_j \}^T \\ \sum_{i=1}^r h_i(p(t)) \mathbf{E}_i^T \\ \mathbf{P}^{-1} \end{array} \right] \geq \mathbf{0}. \quad (36)$$

The left-hand side of (36) can be rewritten as

$$\left[\begin{array}{cc} \mathbf{P} & \mathbf{0} \\ \mathbf{0} & \gamma^2 \mathbf{I} \\ \sum_{i=1}^r \sum_{j=1}^r h_i(p(t)) h_j(p(t)) (\mathbf{A}_i - \mathbf{B}_i \mathbf{F}_j) & \sum_{i=1}^r h_i(p(t)) \mathbf{E}_i \\ \sum_{i=1}^r \sum_{j=1}^r h_i(p(t)) h_j(p(t)) \{ \mathbf{A}_i - \mathbf{B}_i \mathbf{F}_j \}^T & \\ \sum_{i=1}^r h_i(p(t)) \mathbf{E}_i^T & \\ \mathbf{P}^{-1} & \end{array} \right]$$

$$- \left[\begin{array}{c} \sum_{i=1}^r h_i(p(t)) \mathbf{C}_i^T \\ 0 \\ 0 \end{array} \right] \left[\begin{array}{ccc} \sum_{i=1}^r h_i(p(t)) \mathbf{C}_i & 0 & 0 \end{array} \right].$$

Again, by applying Schur complement, we have the following conditions.

$$\left[\begin{array}{cc} \mathbf{P} & \mathbf{0} \\ \mathbf{0} & \gamma^2 \mathbf{I} \\ \sum_{i=1}^r \sum_{j=1}^r h_i(p(t)) h_j(p(t)) (\mathbf{A}_i - \mathbf{B}_i \mathbf{F}_j) & \sum_{i=1}^r h_i(p(t)) \mathbf{E}_i \\ \sum_{i=1}^r h_i(p(t)) \mathbf{C}_i & \mathbf{0} \\ \sum_{i=1}^r \sum_{j=1}^r h_i(p(t)) h_j(p(t)) (\mathbf{A}_i - \mathbf{B}_i \mathbf{F}_j)^T & \sum_{i=1}^r h_i(p(t)) \mathbf{C}_i^T \\ \sum_{i=1}^r h_i(p(t)) \mathbf{E}_i^T & \mathbf{0} \\ \mathbf{P}^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{array} \right]$$

$$= \sum_{i=1}^r \sum_{j=1}^r h_i(p(t)) h_j(p(t)) \left[\begin{array}{cc} \mathbf{P} & \mathbf{0} \\ \mathbf{0} & \gamma^2 \mathbf{I} \\ \frac{1}{2} (\mathbf{A}_i - \mathbf{B}_i \mathbf{F}_j + \mathbf{A}_j - \mathbf{B}_j \mathbf{F}_i) & \frac{1}{2} (\mathbf{E}_i + \mathbf{E}_j) \\ \frac{1}{2} (\mathbf{C}_i + \mathbf{C}_j) & \mathbf{0} \end{array} \right]$$

$$\begin{bmatrix} \frac{1}{2}(\mathbf{A}_i - \mathbf{B}_i \mathbf{F}_j + \mathbf{A}_j - \mathbf{B}_j \mathbf{F}_i)^T & \frac{1}{2}(\mathbf{C}_i + \mathbf{C}_j)^T \\ \frac{1}{2}(\mathbf{E}_i + \mathbf{E}_j)^T & \mathbf{0} \\ \mathbf{P}^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{bmatrix} \geq \mathbf{0} \quad (37)$$

Therefore,

$$\begin{bmatrix} \mathbf{P} & \mathbf{0} \\ \mathbf{0} & \gamma^2 \mathbf{I} \\ \frac{1}{2}(\mathbf{A}_i - \mathbf{B}_i \mathbf{F}_j + \mathbf{A}_j - \mathbf{B}_j \mathbf{F}_i) & \frac{1}{2}(\mathbf{E}_i + \mathbf{E}_j) \\ \frac{1}{2}(\mathbf{C}_i + \mathbf{C}_j) & \mathbf{0} \\ \frac{1}{2}(\mathbf{A}_i - \mathbf{B}_i \mathbf{F}_j + \mathbf{A}_j - \mathbf{B}_j \mathbf{F}_i)^T & \frac{1}{2}(\mathbf{C}_i + \mathbf{C}_j)^T \\ \frac{1}{2}(\mathbf{E}_i + \mathbf{E}_j)^T & \mathbf{0} \\ \mathbf{P}^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{bmatrix} \geq \mathbf{0}. \quad i \leq j \quad (38)$$

By multiplying both sides of (38) by $block-diag[\mathbf{X} \ \mathbf{I} \ \mathbf{I} \ \mathbf{I}]$, where $\mathbf{X} = \mathbf{P}^{-1}$, (32) is obtained.

(Q.E.D.)

[Remark 3.1]

An advantage of our approach for LPV control, e.g., [31]-[33], is relaxation of conditions by utilizing some permutation techniques [34].

4 Simulation Results

In applying the LMI-based fuzzy control design to the backing up control of a vehicle with triple trailers, we investigate design conditions involving stability, decay rate, constraints on the input and constraints on the output and disturbance rejection.

The purpose of considering decay rate is to achieve a desired rate of backing up into the straight line. The system settles on to the straight line quicker for a larger decay rate. However, an aggressive decay rate could result in the occurrence of the jack-knife phenomenon and the saturation of the steering angle.

The control input is the steering angle of the vehicle. The objective of the input constraint is to avoid the saturation of the steering angle.

The output is the relative angles between the vehicle and the first trailer, the first trailer and the second trailer, and the second trailer and the third trailer. The purpose of the output constraints is to avoid the jack-knife phenomenon.

The following design parameters are used in the simulation:

- The constraint on the input is $\mu = 15$ [deg.].
- The constraints on the outputs are $\lambda_i = 90$ [deg.] for $i = 1, 2, 3$.

The control input constraint " $\mu = 15$ [deg.]" is the limitation of the steering angle of our vehicle. " $\lambda = 90$ [deg.]" directly means the avoidance of jack-knife phenomenon. Figure 4 shows the simulation results of an easy initial position for the stable fuzzy controller and the decay rate fuzzy controller. Figure 5 shows the simulation results of a difficult initial position for the stable fuzzy controller, the decay rate fuzzy controller and the fuzzy controller satisfying decay rate and constraint on control input and output. The following important remarks can be made from the simulation results.

[Remark 4.1]

When we only invoke the stability conditions in the design, the closed-loop system does not necessarily have the desired performances in terms of decay rate and other specifications. Decay rate condition is included in the design to arrive at a speedy response of the controller system.

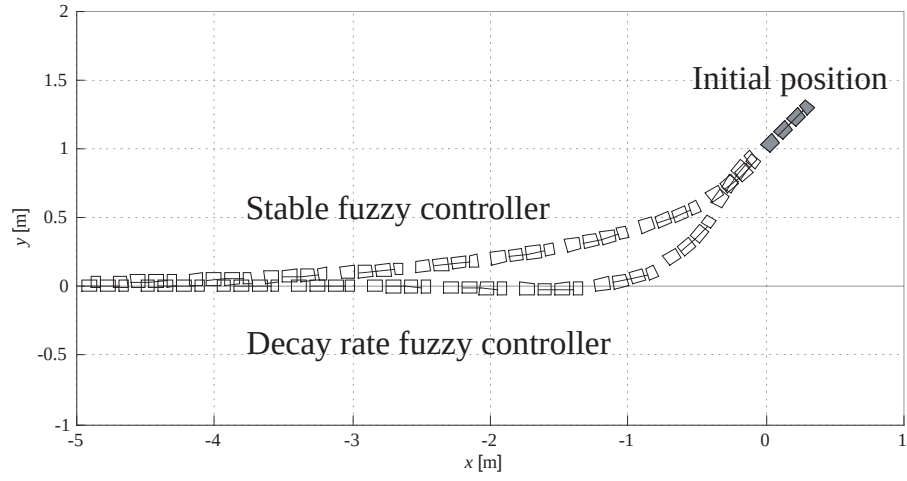


Figure 4: Simulation result 1.

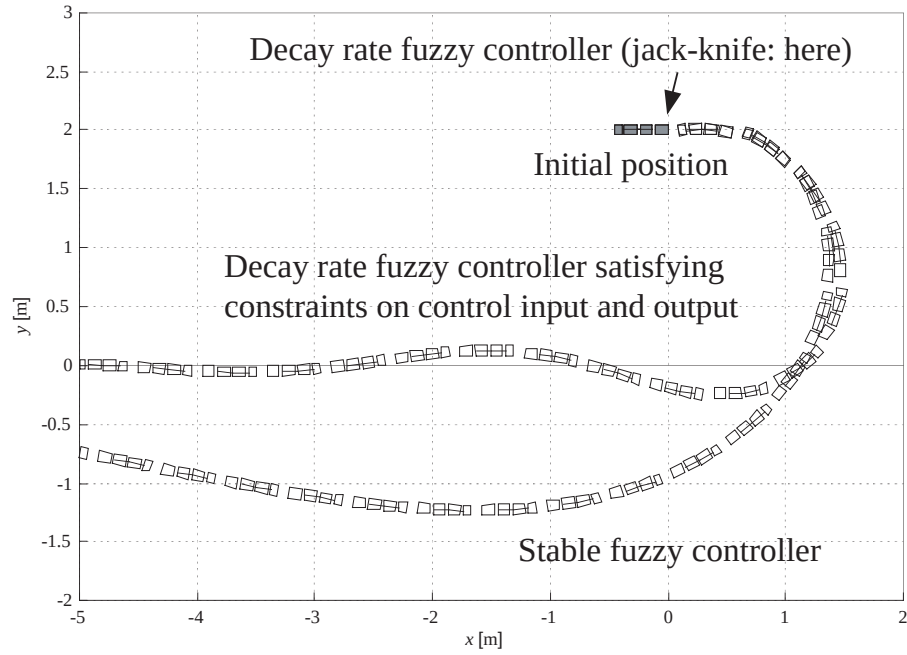


Figure 5: Simulation result 2.

[Remark 4.2]

When the vehicle is at an "easy" initial position, the decay rate design is effective, i.e., the vehicle approaches the desired straight line quickly. However, the vehicle starts from a "difficulty" initial position, the following some problems occur. The first problem is the occurrence of the saturation of the steering angle. The second problem is the occurrence of the "jack-knife" phenomenon. In Figure 5, the decay rate control occurs jack-knife phenomenon as soon as control starts.

[Remark 4.3]

To circumvent these problems, we invoke design conditions involving input constraints (avoiding the steering angle saturation), output constraints (avoiding jack-knife phenomenon) in addition to stability and decay rate. Figure 6 shows the effect of constraint on the control input. The steering angle of the controller considering the input constraint is within ± 15 [deg.].

Next, we show the effect of disturbance rejection. Figure 7 shows control result for the disturbance $\mathbf{v}(t) = \frac{8\pi}{180} \sin(t)[rad.]$, where

$$\mathbf{E}_i = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{C}_i = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix},$$

for $i = 1, 2$. This means that $\mathbf{v}(t)$'s are added to the angles $\theta_{01}(t)$, $\theta_{12}(t)$ and $\theta_{23}(t)$ of each connection, where the maximum values of each element in $\mathbf{v}(t)$ correspond to $\pm 8[deg.]$. The decay rate fuzzy controller could no longer avoid jack-knife phenomenon. The other decay

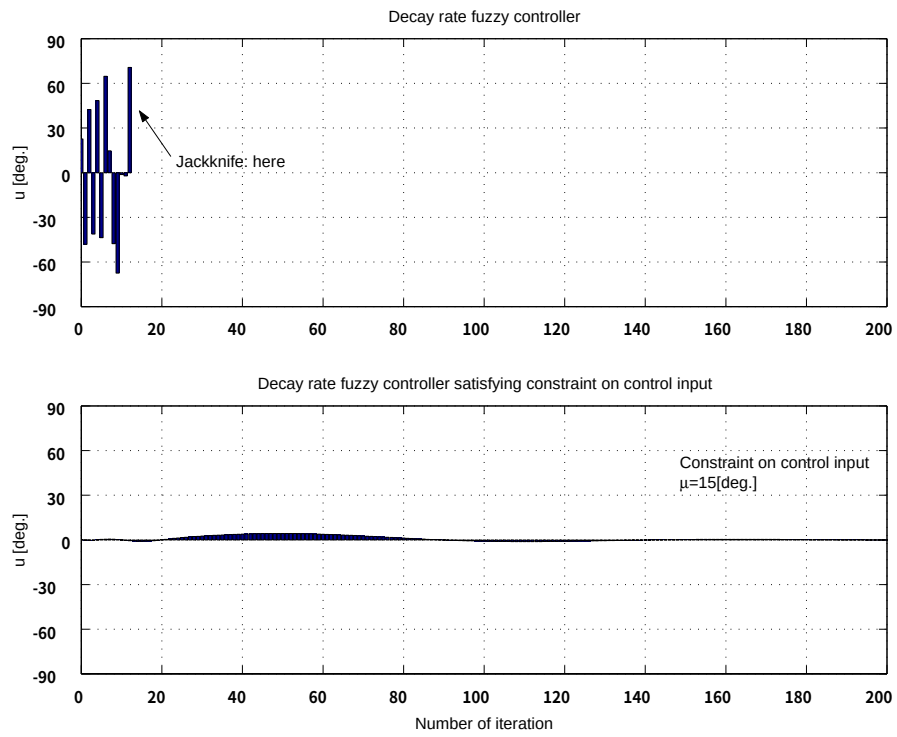


Figure 6: Effect of constraint on control input.

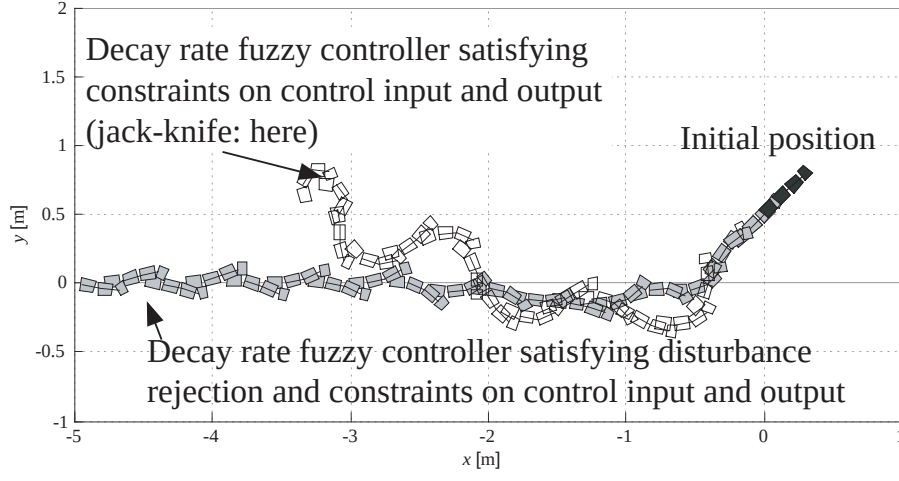


Figure 7: Simulation result 3.

rate fuzzy controller that considers disturbance rejection succeeds in the backing control although the trailer oscillates around $y(t)$ due to a large disturbance.

Figure 8 shows control result for a larger disturbance $\mathbf{v}(t) = \frac{10\pi}{180} \sin(t)[rad.]$, where

$$\mathbf{E}_i = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \mathbf{C}_i = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix},$$

for $i = 1, 2$. Figure 9 shows the magnified area (area A in Figure 8) around the initial positions. The decay rate fuzzy controller that considers disturbance rejection works well even for the large disturbance.

5 Experimental Results

Figure 10 shows the photograph of the vehicle with triple trailers. Figure 11 shows the experimental system. The connecting angles $\theta_{01}(t)$, $\theta_{12}(t)$ and $\theta_{23}(t)$ are observed by three potentiometers. A CCD camera is used to detect the x-y position of the third trailer. Since

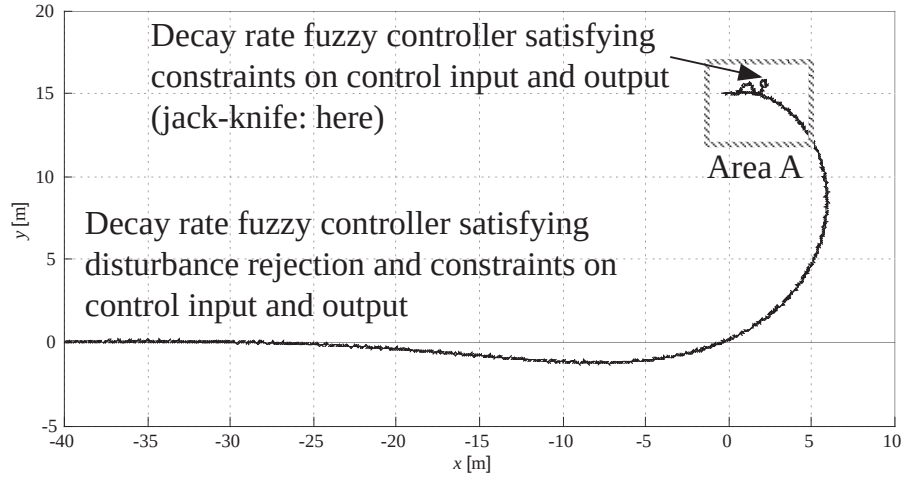


Figure 8: Simulation result 4.

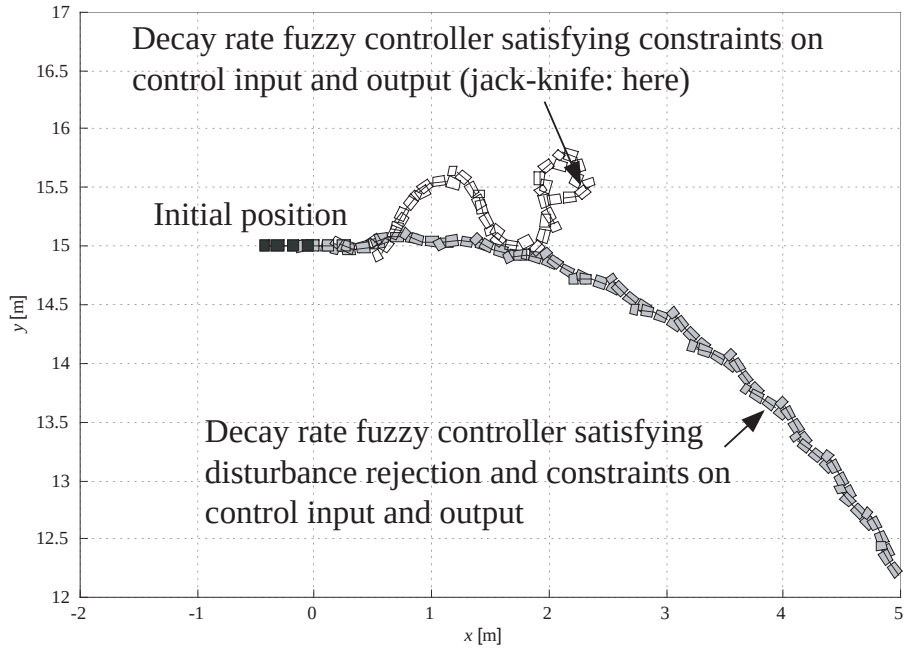


Figure 9: Magnification of figure 8 (Area A).

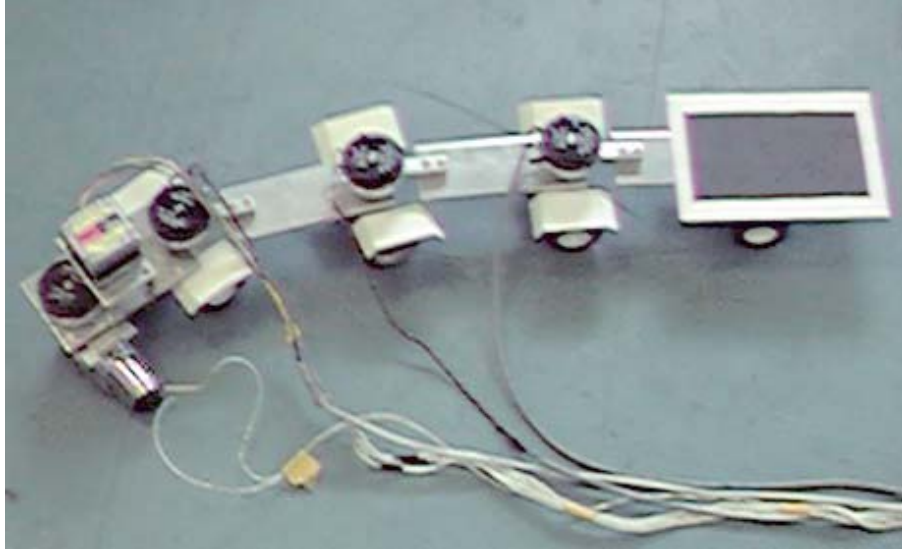


Figure 10: Photograph of vehicle with triple trailers.

the processing time for detecting the x-y position is long, the vehicle is stopped during the position detection at every sampling. After the position detection, the steering is controlled by a stepping motor and then the vehicle is shortly moved by a DC motor. The moving distance at every sampling is $0.05[m]$. Due to the short movement and stop sequence, we assume that the mass and inertia may be ignored. Although we use a simple kinematic-like model (39) - (46) with the assumptions, our experimental results show that a fuzzy controller designed from the simple model works well in real experiments.

Figure 12 and Figure 13 show experimental results. It is found from these figures that backward movement control of the vehicle with triple trailers can be effectively realized by the fuzzy controller.

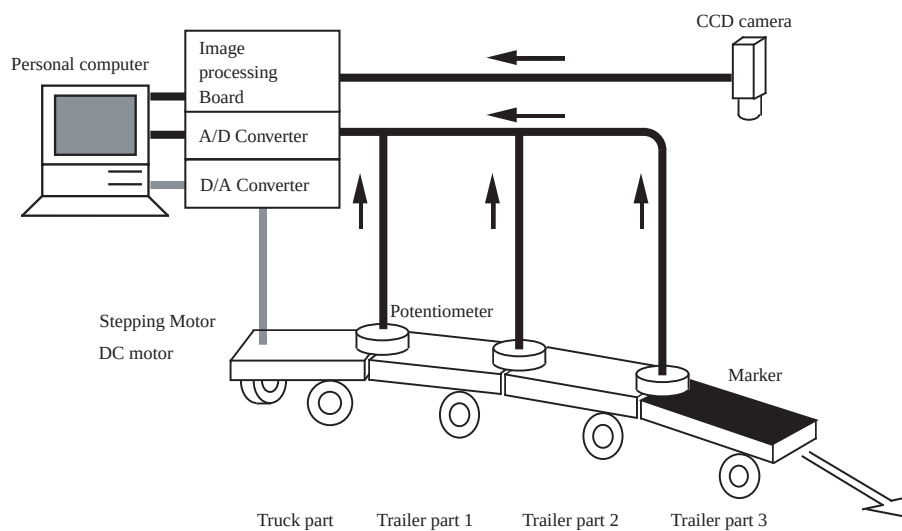


Figure 11: Experimental system.

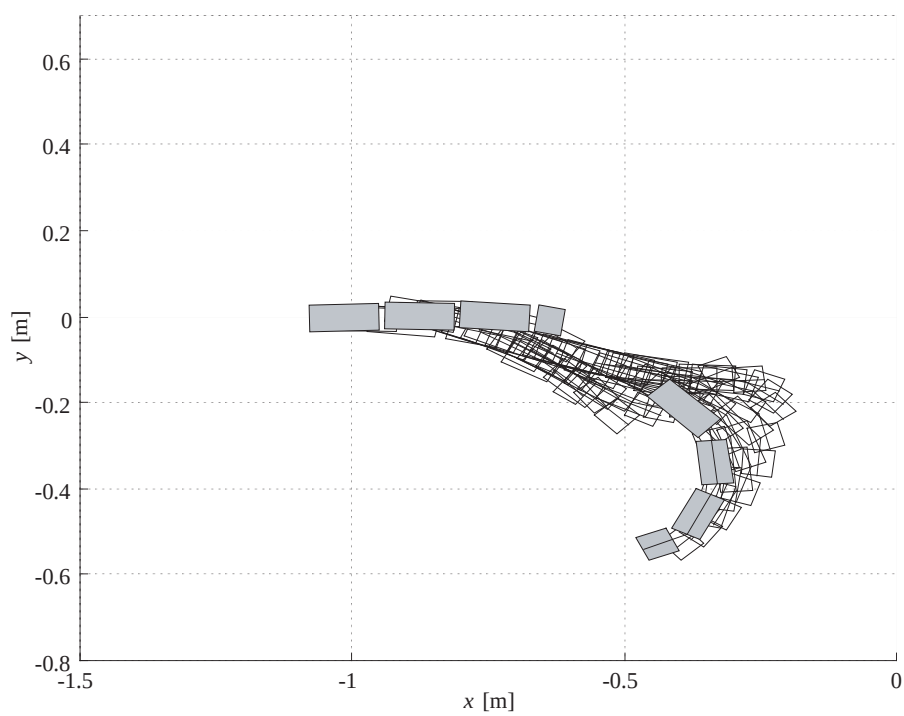


Figure 12: Experimental result 1.

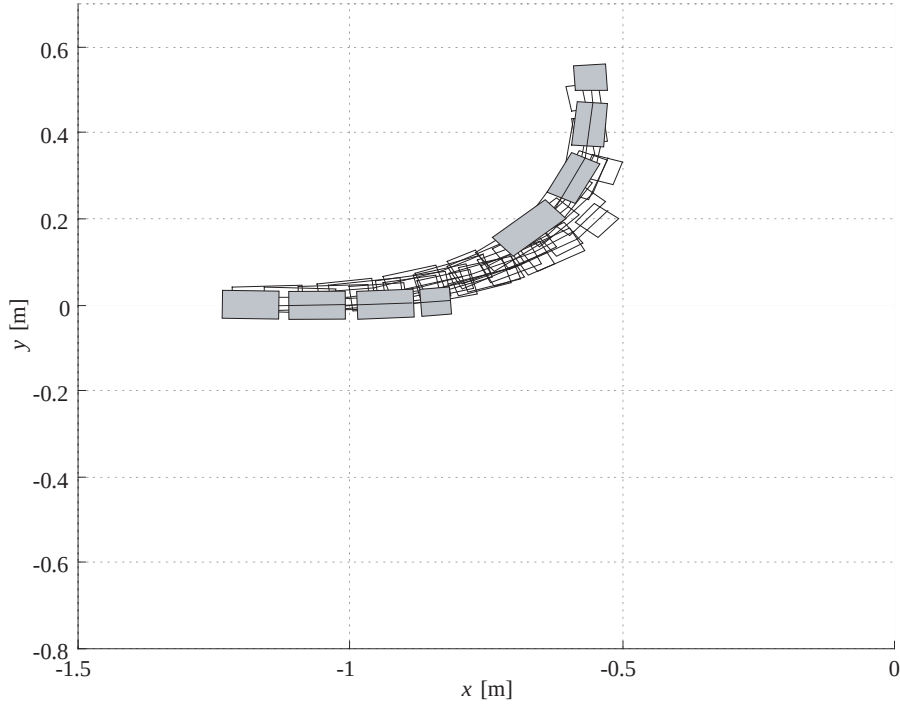


Figure 13: Experimental result 2.

6 Conclusion

We have designed a control system for the backing up control of a vehicle with triple trailers. The design has employed a model-based fuzzy control methodology. A Takagi-Sugeno fuzzy model is constructed to describe the nonlinear dynamics of the vehicle. LMI (linear matrix inequality) based fuzzy control design conditions have been derived in the framework of the parallel distributed compensation and the relaxed stability conditions. This paper has presented not only stability but also performance specifications in terms of decay rate, constraints on the control input, constraints on the output and disturbance rejection. These LMI conditions together achieve multi-objective control design which entails quick settling of the system, avoidance of the steering angle saturation and the jack-knife phenomenon.

The design procedure is simple and systematic, in contrast to typical trial-and-error fuzzy control design. Simulation results and experimental results have demonstrated the validity and effectiveness of the fuzzy control methodology presented in this paper.

Appendix Fuzzy modeling

To succeed in the backing control experiments, we try to simplify the nonlinear equations (1)-(7) as much as possible. The simplification is useful for considering many practical constraints on sensors, actuators, etc. In our experience through the experiments [11], the backing control never succeed when $u(t)$, $\theta_{01}(t)$, $\theta_{12}(t)$ and $\theta_{23}(t)$ are not small. The variables are strongly related to "jack-knife". To facilitate the model simplification and control design, with the assumption that the values of $u(t)$, $\theta_{01}(t)$, $\theta_{12}(t)$ and $\theta_{23}(t)$ are not large for all times, we obtain the following simplified model of the system:

$$\theta_0(t+1) = \theta_0(t) + \frac{\nu \cdot \Delta t}{\ell} u(t), \quad (39)$$

$$\theta_{01}(t+1) = \left(1 - \frac{\nu \cdot \Delta t}{L}\right) \theta_{01}(t) + \frac{\nu \cdot \Delta t}{\ell} u(t), \quad (40)$$

$$\theta_1(t+1) = \theta_1(t) + \frac{\nu \cdot \Delta t}{L} \theta_{01}(t), \quad (41)$$

$$\theta_{12}(t+1) = \left(1 - \frac{\nu \cdot \Delta t}{L}\right) \theta_{12}(t) + \frac{\nu \cdot \Delta t}{L} \theta_{01}(t), \quad (42)$$

$$\theta_2(t+1) = \theta_2(t) + \frac{\nu \cdot \Delta t}{L} \theta_{12}(t), \quad (43)$$

$$\theta_{23}(t+1) = \left(1 - \frac{\nu \cdot \Delta t}{L}\right) \theta_{23}(t) + \frac{\nu \cdot \Delta t}{L} \theta_{12}(t), \quad (44)$$

$$\theta_3(t+1) = \theta_3(t) + \frac{\nu \cdot \Delta t}{L} \theta_{23}(t), \quad (45)$$

$$y(t+1) = y(t) + \nu \cdot \Delta t \sin \left(\theta_3(t) + \frac{\nu \cdot \Delta t}{2L} \theta_{23}(t) \right). \quad (46)$$

Note that (46) has the nonlinear term,

$$\nu \cdot \Delta t \cdot \sin \left(\theta_3(t) + \frac{\nu \cdot \Delta t}{2L} \theta_{23}(t) \right). \quad (47)$$

This nonlinear term can be represented by the following Takagi-Sugeno fuzzy model representation,

$$\begin{aligned} & \nu \cdot \Delta t \cdot \sin \left(\theta_3(t) + \frac{\nu \cdot \Delta t}{2L} \theta_{23}(t) \right) \\ = & w_1(p(t)) \cdot \nu \cdot \Delta t \cdot \left(\theta_3(t) + \frac{\nu \cdot \Delta t}{2L} \theta_{23}(t) \right) \\ & + w_2(p(t)) \cdot \nu \Delta t \cdot g \cdot \left(\theta_3(t) + \frac{\nu \cdot \Delta t}{2L} \theta_{23}(t) \right), \end{aligned} \quad (48)$$

where

$$p(t) = \theta_3(t) + \frac{\nu \cdot \Delta t}{2L} \theta_{23}(t),$$

$$g = 10^{-2}/\pi,$$

$$w_1(p(t)) = \begin{cases} \frac{\sin(p(t)) - g \cdot p(t)}{p(t) \cdot (1 - g)}, & p(t) \neq 0 \\ 1 & p(t) = 0 \end{cases} \quad (49)$$

$$w_2(p(t)) = \begin{cases} \frac{p(t) - g \cdot \sin(p(t))}{p(t) \cdot (1 - g)}, & p(t) \neq 0 \\ 1 & p(t) = 0 \end{cases} \quad (50)$$

From (49) and (50), it can be seen that $w_1(p(t)) = 1$ and $w_2(p(t)) = 0$ when $p(t)$ is about 0 [rad.]. Similarly, $w_1(p(t)) = 0$ and $w_2(p(t)) = 1$ when $p(t)$ is about π or $-\pi$ [rad.].

When $w_1(p(t)) = 1$ and $w_2(p(t)) = 0$, i.e., $p(t)$ is about 0 [rad.], substituting (48) into (46), we have

$$y(t+1) = y(t) + \nu \cdot \Delta t \cdot \theta_3(t) + \frac{(\nu \cdot \Delta t)^2}{2L} \cdot \theta_{23}(t).$$

Therefore the simplified nonlinear model can be represented by

$$\begin{aligned}
\begin{bmatrix} \theta_{01}(t+1) \\ \theta_{12}(t+1) \\ \theta_{23}(t+1) \\ \theta_3(t+1) \\ y(t+1) \end{bmatrix} &= \begin{bmatrix} 1 - \frac{\nu \cdot \Delta t}{L} & 0 & 0 & 0 & 0 \\ \frac{\nu \cdot \Delta t}{L} & 1 - \frac{\nu \cdot \Delta t}{L} & 0 & 0 & 0 \\ 0 & \frac{\nu \cdot \Delta t}{L} & 1 - \frac{\nu \cdot \Delta t}{L} & 0 & 0 \\ 0 & 0 & \frac{\nu \cdot \Delta t}{L} & 1 & 0 \\ 0 & 0 & \frac{(\nu \cdot \Delta t)^2}{2L} & \nu \cdot \Delta t & 1 \end{bmatrix} \begin{bmatrix} \theta_{01}(t) \\ \theta_{12}(t) \\ \theta_{23}(t) \\ \theta_3(t) \\ y(t) \end{bmatrix} \\
&+ \begin{bmatrix} \frac{\nu \cdot \Delta t}{\ell} \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} u(t). \tag{51}
\end{aligned}$$

When $w_1(p(t)) = 1$ and $w_2(p(t)) = 0$, i.e., $p(t)$ is about π or $-\pi$ [deg.], (46) is reduced to

$$y(t+1) = y(t) + g \cdot \nu \cdot \Delta t \cdot \theta_3(t) + \frac{g \cdot (\nu \cdot \Delta t)^2}{2L} \cdot \theta_{23}(t).$$

Therefore the simplified nonlinear model can be represented by

$$\begin{aligned}
\begin{bmatrix} \theta_{01}(t+1) \\ \theta_{12}(t+1) \\ \theta_{23}(t+1) \\ \theta_3(t+1) \\ y(t+1) \end{bmatrix} &= \begin{bmatrix} 1 - \frac{\nu \cdot \Delta t}{L} & 0 & 0 & 0 & 0 \\ \frac{\nu \cdot \Delta t}{L} & 1 - \frac{\nu \cdot \Delta t}{L} & 0 & 0 & 0 \\ 0 & \frac{\nu \cdot \Delta t}{L} & 1 - \frac{\nu \cdot \Delta t}{L} & 0 & 0 \\ 0 & 0 & \frac{\nu \cdot \Delta t}{L} & 1 & 0 \\ 0 & 0 & \frac{g \cdot (\nu \cdot \Delta t)^2}{2L} & g \cdot \nu \cdot \Delta t & 1 \end{bmatrix} \begin{bmatrix} \theta_{01}(t) \\ \theta_{12}(t) \\ \theta_{23}(t) \\ \theta_3(t) \\ y(t) \end{bmatrix} \\
&+ \begin{bmatrix} \frac{\nu \cdot \Delta t}{\ell} \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} u(t). \tag{52}
\end{aligned}$$

In this representation, if $g = 0$, the system (52) becomes uncontrollable. To handle the problem, we select $g = 10^{-2}/\pi$. With this choice of g , the nonlinear term of (47) is exactly represented by the expression of (48) under the condition

$$-179.4270[\text{deg.}] < p(t) < 179.4270[\text{deg.}].$$

From (51) and (52), we arrive at the Takagi-Sugeno fuzzy model (10).

References

- [1] O. J. Sørndalen, "Conversion of the kinematics of a car with n trailers into chained form", *Proc. IEEE Int. Conf. on Robotics and Automation*, pp.382-387, Atlanta, Georgia, (1993).
- [2] C. Altafini, "Controllability and singularities in the n-trailer system with kingpin hitching", *Proc. 14th IFAC World Congress*, vol.Q, pp.139-145, Beijing, China, (1999).
- [3] A. W. Divelbiss and J. Wen, "Nonholonomic path planning with inequality constraints", *Proc. IEEE Int. Conf. Robotics and Automation*, pp.52-56, (1994).
- [4] D. Nguyen and B. Widrow, "The Truck Backer-upper :An Example of Self-Learning in Neural Networks", *Proc. Int. Joint Conf. Neural Networks(IJCNN-89)*, vol.2, pp.357-363, (1989).
- [5] G.S. Kong and B. Kosko, "Adaptive Fuzzy Systems for Backing up a Truck-and-Trailer", *IEEE Trans. on Neural Networks*, vol.3, no.2, pp. 211-223, (1992).
- [6] H.Inoue, K.Kamei and K.Inoue, "Auto-Generation of Fuzzy Production Rules Using Hyper-Cone Membership Function by Genetic Algorithm", *Proc. Int. Joint Conf. CFSA/IFIS/SOFT'95*, pp.53-58, (1995).
- [7] M. Tokunaga and H. Ichihashi, "Backer-Upper Control of a Trailer Truck by Neuro-Fuzzy Optimal Control", *Proc. of 8th Fuzzy System Symposium*, pp.49-52, in Japanese (1992).

- [8] Chen Guanrong and Zhang Delin, "Back-driving a truck with suboptimal distance trajectories: a fuzzy logic control approach", *IEEE Trans. on Fuzzy Systems*, vol.3, no.1, pp.125-131, (1995).
- [9] M. Sampei, et al., "Arbitrary Path Tracking Control of Articulated Vehicles Using Nonlinear Control Theory", *IEEE Trans. on Control Systems Technology*, vol. 5, no. 3, pp. 369-80, Aug. (1997).
- [10] M. Sampei, T. Kobayashi, Y. Tajima, N. Shibui and M. Ishikawa, "Application of Nonlinear Control Theory to Path Tracking Control of Articulated Vehicles", *1995 IEEE International Conference on Robotics and Automation, Video Proceedings* (1995).
- [11] K.Tanaka and T. Kosaki, "Design of a Stable Fuzzy Controller for an Articulated Vehicle", *IEEE Transactions on Systems, Man and Cybernetics : Part B*, Vol.27, No.3, pp.552-558 (1997).
- [12] K. Tanaka and M. Sano, "Stability Analysis of Fuzzy Systems Using Lyapunov's Direct Method", *Proc. of NAFIPS'90* , pp.133-136, (1990).
- [13] K. Tanaka, and M.Sugeno,"Stability Analysis and Design of Fuzzy Control Systems," *FUZZY SETS AND SYSTEMS*, vol.45, no.2, pp.135-156, (1992).
- [14] H. O. Wang, K. Tanaka and M. Griffin, "Parallel Distributed Compensation of Fuzzy Systems via Takagi-Sugeno Fuzzy Model", *Proc. of 4th IEEE Int. Conf. on Fuzzy Systems* vol.2, pp.531-538, (1995).

- [15] H. O. Wang, K. Tanaka and M. Griffin, "An Approach to Fuzzy Control of Nonlinear Systems", *IEEE Transactions on Fuzzy Systems*, vol.4, No.1, pp.14-23, (1996).
- [16] H. O. Wang and K. Tanaka, "An Analytical Framework of Fuzzy Modeling and Control of Nonlinear Systems: Stability and Design Issues", *Proc. of 1995 American Control Conference*, vol.3, pp.2272-2276, (1995).
- [17] Y. Z. Lu, M. He and C. W. Xu, "Fuzzy Modeling and Expert Optimization Control for Industrial Processes", *IEEE Transactions on Control Systems Technology*, vol.5, no.1, pp.2-12, (1997).
- [18] H. A. Malki, D. Misir, D. Feigenspan and G. Chen, "Fuzzy PID Control of a Flexible-Joint Robot Arm with Uncertainties from Time-Varying Loads", *IEEE Transactions on Control Systems Technology*, vol.5, no.3, pp.371-378, (1997).
- [19] L. C. Lin and T. B. Gau, "Feedback Linearization and Fuzzy Control for Conical Magnetic Bearings", *IEEE Transactions on Control Systems Technology*, vol.5, no.4 pp.417-426, (1997).
- [20] M. Dussud, S. Galichet and L. P. Foulloy, "Application of Fuzzy Logic Control for Continuous Casting Mold Level Control", *IEEE Transactions on Control Systems Technology*, vol.6, no.2 pp.246-256, (1998).
- [21] R. Langari and M. Tomizuka, "Analysis and Synthesis of Fuzzy Linguistic Control Systems," *1990 ASME Winter Annual Meeting*, pp. 35-42, (1990).

- [22] T. Kosaki et. al., "Model-Based Fuzzy Control System Design for Magnetic Bearings", *Proc. of Sixth IEEE International Conference on Fuzzy Systems*, Barcelona, Vol.2, pp.895-899 (1997).
- [23] S. S. Farinwata et. al., "Stability Analysis of The Fuzzy Logic Controller Designed by The Phase Portrait Assignment Algorithm", *Proc. of 2nd IEEE International Conference on Fuzzy Systems*, pp.1377-1382, (1993)
- [24] D. Filev, "Algebraic Design of Fuzzy Logic Controllers", *Proc. of 1996 IEEE International Symposium on Intelligent Control*, pp.253-258, (1996)
- [25] S. Masui et.al., "Man-machine System for Triple Inverted Pendulum by Fuzzy Logic and Genetic Algorithm," *Proc. of IFSA'97*, vol.2, pp.398-403, (1997).
- [26] T. Furuhashi et.al., "A Stability Analysis of Fuzzy Control System Using a Generalized Fuzzy Petri Net Model", *Proc. of Seventh IEEE International Conference on Fuzzy Systems*, pp.95-100, (1998)
- [27] S. Farinwata and G. Vachtsevanos, "Robust Stability of Fuzzy Logic Controller Systems," *Proc. of 1995 American Control Conference*, pp2267-2271, (1995).
- [28] D. Filev and P. Angelov, "Fuzzy Optimal Control," *Fuzzy Sets and Systems*, 47, pp.151-156, (1992).
- [29] T. Takagi and M. Sugeno,"Fuzzy Identification of Systems and Its Applications to Modeling and Control," *IEEE SMC*, vol.15, no.1, pp.116-132, (1985).

- [30] K. Tanaka, T. Ikeda and H. O. Wang, "Fuzzy Regulators and Fuzzy Observers: Relaxed Stability Conditions and LMI-Based Designs", *IEEE transactions on fuzzy systems*, Vol.6 No.2, pp.250-265, May. (1998).
- [31] H. Kajiwarara et.al., "LPV Techniques for Control of an Inverted Pendulum", *IEEE Control Systems Magazine*, vol.19, no.1, pp.44-54 (1999).
- [32] P. Apkarian "On the Discretization of LMI-Synthesized Linear Parameter-Varying Controllers", *Automatica*, vol. 33, no. 4, pp. 655-661, (1997).
- [33] P. Gahinet et.al., "Affine parameter-dependent Lyapunov functions and real parametric uncertainty", *IEEE Transactions on Automatic Control*, vol.41, no.3, pp.436-442 (1996).
- [34] K. Tanaka and H. O. Wang, "Fuzzy Control Systems Design and Analysis", *JOHN WILEY & SONS, INC.* (2001).

Figure caption list

- Figure 1 Vehicle model with triple trailers
- Figure 2 Membership function
- Figure 3 Jack-knife phenomenon
- Figure 4 Simulation result 1
- Figure 5 Simulation result 2
- Figure 6 Effect of constraint on control input
- Figure 7 Simulation result 3
- Figure 8 Simulation result 4
- Figure 9 Magnification of figure 8
- Figure 10 Photograph of vehicle with triple trailers
- Figure 11 Experimental system
- Figure 12 Experimental result 1
- Figure 13 Experimental result 2